



Proposed Capstone Project Ideas

by
mTechPesOct24AGrp9

Our Team

Divya Narasimha Prasanna
Priyanka Jain
Rajani M S
Sunaina Kouser Shukoor
Upadhyay Kiran Maheshchandra





Idea 1

AI-based Defect De-Duplication System



Objectives

Our aim

- To design an AI-based system capable of detecting duplicate defect reports.
- To apply NLP techniques to understand semantic similarity between defect descriptions.
- To implement machine learning / deep learning models for accurate classification.
- To reduce manual triaging effort and improve bug resolution time.
- To build a scalable and deployable prototype for real-world usage.

The goal

- Automated identification of duplicate defects with high accuracy (>85%).
- Reduction in manual triaging time by 60–80%.
- Faster bug resolution cycles.
- Scalable prototype suitable for enterprise integration.

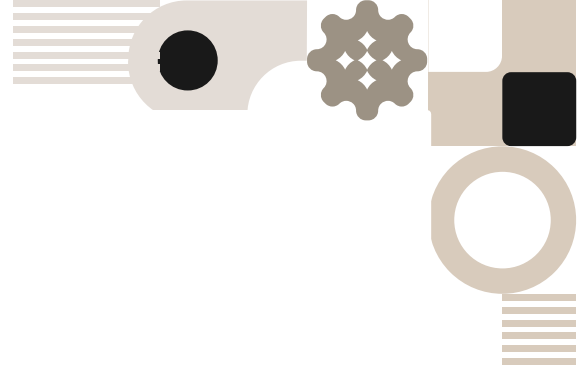
Scope

The project scope includes:

- Preprocessing and cleaning of defect report text data.
- Feature extraction using traditional NLP and transformer-based embeddings.
- Training supervised and unsupervised ML models for similarity detection.
- Development of a REST-based or web-based interface for duplicate prediction.
- Performance evaluation using standard NLP metrics.

Out of scope:

- Automated bug fixing.
- Root cause analysis automation.



Idea 2

Selection, Optimization, and Performance Analysis of Transformer- Based Language Models for Text Classification



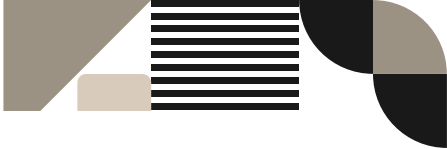
Problem Statement & Gap

Problem Statement:

- Multiple transformer-based language models are available
- Organizations struggle to decide:
 - Which model to use?
 - How much fine-tuning is needed?
 - What is the trade-off between accuracy and cost?
 - Most projects focus on **one model**, not **model selection**

Gap:

- Lack of a systematic comparison framework for model selection under real-world constraints.



Proposed Idea & Use Case

What We Will Do:

- Compare multiple transformer-based models:
 - DistilBERT
 - BERT
 - RoBERTa
 - Apply them to text classification tasks
- Evaluate using:
- Accuracy metrics
 - Training efficiency
 - Inference latency

Outcome: Guidelines to select the most suitable model for deployment.

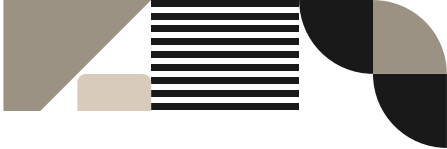
Primary Use Case:

- Text Classification (foundational NLP task)

Applicable Domains:

- Support ticket classification
- Sentiment analysis
- Document routing systems
- News categorization

The project is domain-agnostic and focuses on model behavior rather than one application.



Data, Techniques & Evaluation

Datasets (Real-World):

- **AG News :**
 - Real news articles
 - Multi-class classification
- **IMDb Movie Reviews :**
 - Real user-written reviews
 - Binary sentiment classification

Why these datasets?:

- Public, real-world, and widely used
- Support reproducibility
- Cover different classification settings

Models:

- DistilBERT (lightweight)
- BERT-base
- RoBERTa-base

Techniques:

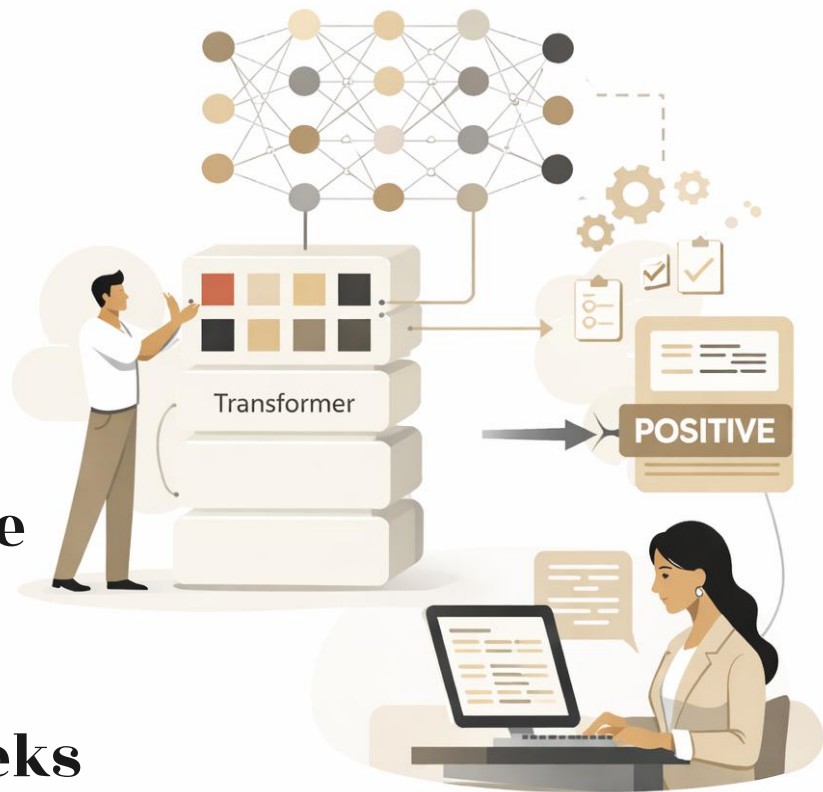
- Fine-tuning
- Partial layer freezing
- Optimizer and learning rate comparison

Evaluation Metrics:

- **Performance Metrics:** Accuracy, Precision, Recall, F1-score
- **Efficiency Metrics:** Training time, GPU memory usage
- **Inference Metrics:** Inference latency

Idea 3

An AI agent that helps clinical researchers quickly navigate unstructured medical literature and case studies, delivering evidence-based insights and reducing review time from weeks to days





Problem

Problem

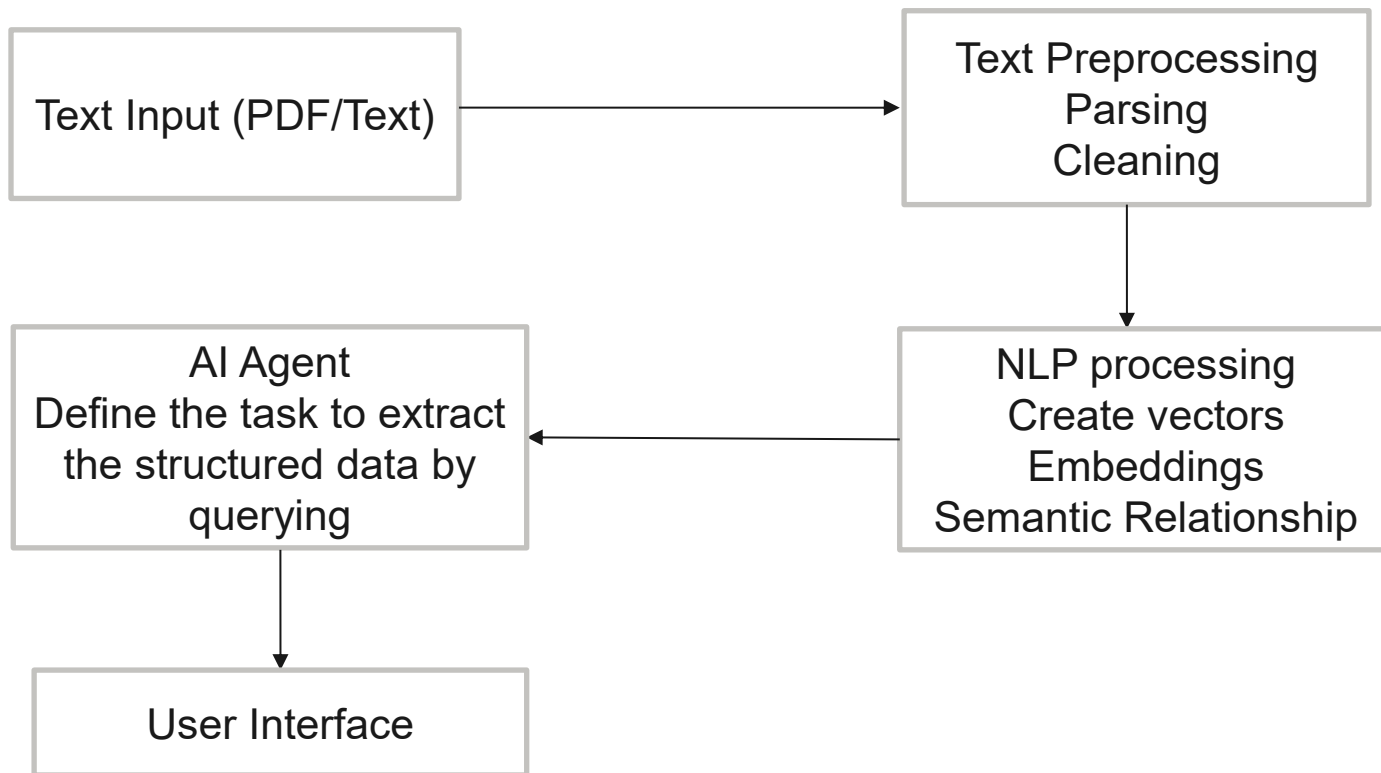
Clinical researchers spend **weeks manually reviewing**:

- Unstructured medical literature
- Case studies
- Clinical trial reports
- PDFs, guidelines, EMRs, and white papers

This leads to:

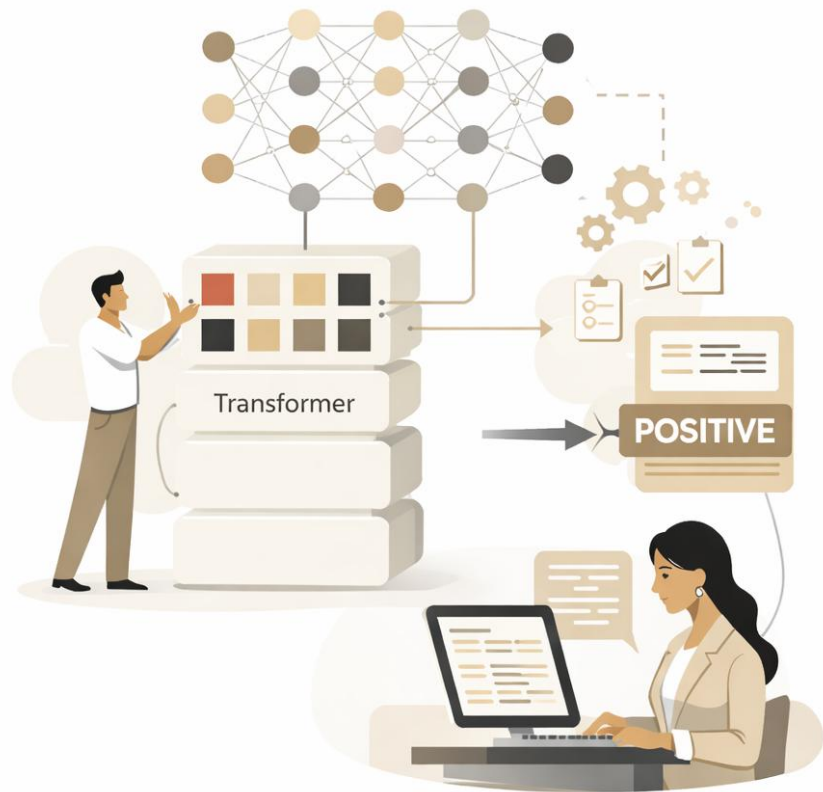
- Slow evidence synthesis
- Information overload
- Missed insights
- High human effort & cost

Solution



Idea 4

Agentic-EdTech Personalized AI Tutor with Long-Term Memory





Problem:

Most existing e-learning platforms provide static or shallow personalization. They adapt content only based on recent quizzes or session-level performance,

Ignoring:

- Long-term learning behavior
- Forgetting patterns
- Learning speed differences
- Concept dependencies

This results in one-size-fits-all learning, poor retention, and disengagement.

Solution:

This project proposes an **Agentic AI-based Personalized Tutor** that:

- Acts as an autonomous tutor agent
- Observes student interactions continuously
- Maintains long-term memory of
 1. Strengths & weaknesses
 2. Learning pace
 3. Mistake patterns
 4. Concept mastery over time
- Plans personalized learning paths
- Adapts content difficulty in real time
- Learns from feedback (student success/failure)

The tutor is not just a recommender — it is an autonomous **decision-making agent**.

Software & Programming Used: The project is implemented primarily in **Python**. **PyTorch** and **Stable-Baselines3** are used to build the reinforcement-learning tutor agent, while **Gymnasium** defines the student–tutor interaction environment. **Pandas/NumPy** handle student datasets and long-term memory storage, with optional **Flask/Streamlit** for a demo interface.

Data Source

We can combine Real+synthetic datasets.

Publicly Available Datasets : ASSISTments Dataset

EdNet Dataset : Large-scale real student data

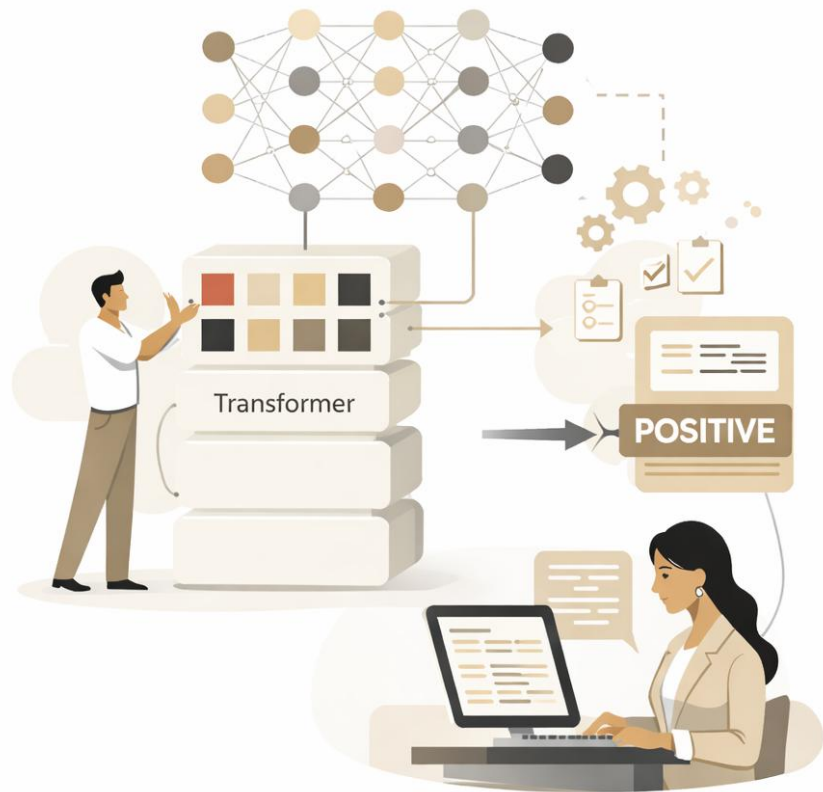
KDD Cup Educational Datasets : ·Student problem-solving sequences

Research Gap:

- Gap 1: Lack of Long-Term Personalization : Most tutoring systems Reset or weakly retain student history and Ignore forgetting and concept decay.
- Gap 2: Limited Agent Autonomy : Use static policies and Do not self-improve teaching strategies
- Gap 3: Poor Adaptation to Learning Pace : Treat all students similarly and Struggle with slow or irregular learners
- Gap 5: Limited Explainability : Black-box tutors reduce trust.

Idea 5

Explainable AI (XAI) for Credit Scoring and Loan Approval





Problem:

"Black box" deep learning models (like XGBoost or Neural Nets) offer high accuracy for credit scoring but lack transparency. This project implements and evaluates **Post-hoc Explainability** methods to ensure loan approval decisions are fair and understandable.

Solution:



1. **Model Training:** Train a complex model (Gradient Boosting or Deep Learning) on a dataset like the **Indian Credit Dataset**
2. **Bias Detection:** Use metrics like **Disparate Impact** or **Demographic Parity** to measure if the model discriminates against a protected group (e.g., age or gender).
3. **Explainability Layer:** Implement **SHAP** (Shapley Additive Explanations) to generate global and local feature importance plots.
4. **Fairness Mitigation:** Apply a pre-processing technique (like **Reweighting**) or in-processing technique (like **Adversarial Debiasing**) to reduce bias.
5. **Dashboard:** Build a dashboard (using **Streamlit** or **Dash**) that visualizes the "Why" behind a rejection (e.g., "Loan rejected because credit history < 2 years").

Thanks!

Our Team

Divya Narasimha Prasanna

Priyanka Jain

Rajani M S

Sunaina Kouser Shukoor

Upadhyay Kiran Maheshchandra

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)

